

Two-Phase DC Charging Method for Improved Efficiency and Zero Torque in Motor-Integrated Charger

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Abstract—Recently, the battery voltage of electric vehicles (EVs) has almost doubled for faster charging. However, it is difficult to charge these EVs with existing low-voltage fast chargers. To solve this issue, boost charging topology using the motor drive system has been introduced and applied to commercial EVs. These motor-integrated chargers are controlled as three-phase interleaved boost converters. Since they use low-zero-sequence inductance for boosting, they could generate larger ripple currents and higher losses. This article analyzes the losses and finds a solution for increasing the charging efficiency and eliminating the torque to avoid rotation. Unlike the conventional method using the zero-sequence inductance, it considers the differential-mode inductance by two-phase conduction of the motor. This results in a significant reduction in current ripple, improving the efficiency. In addition, a position controller is proposed to suppress the rotation explicitly. The feasibility of this work is theoretically assessed, and the experimental results conducted in the reduced scale setup demonstrate the effectiveness of the proposed strategy.

Index Terms—DC charging, efficiency, electric vehicle (EV), integrated charger, interleaved charger, rotation suppression.

I. INTRODUCTION

RECENTLY, the voltages of EV batteries have been doubled to 800 V for driving range extension, cable size reduction, and faster charging [1]. However, the elevated battery voltages exceed the voltage limit of fast dc chargers such as the 135 kW Tesla supercharger, which has a voltage rating of 410 V [2], [3], [4]. Additionally, the output voltages of 1.0–1.2

Received 21 April 2024; revised 9 August 2024; accepted 12 September 2024. Date of publication 15 October 2024; date of current version 3 April 2025. This research was supported by Korea Institute of Marine Science & Technology Promotion (KIMST) funded by the Ministry of Oceans and Fisheries, Republic of Korea under Grant RS-2023-00254688, in part by the BK21 FOUR program of the Education and Research Program for Future ICT Pioneers, Seoul National University in 2024, and in part by the Seoul National University Electric Power Research Institute. (Corresponding author: Jung-Ik Ha.)

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Digital Object Identifier 10.1109/TIE.2024.3468730

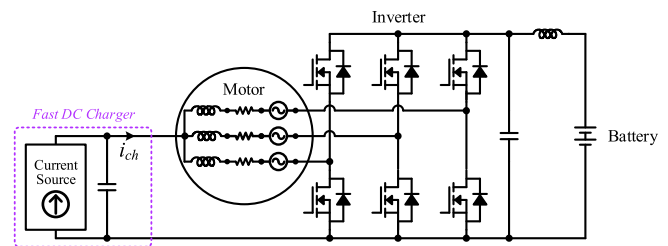


Fig. 1. Schematic of motor-integrated dc charging system utilizing the traction motor windings and a voltage-source inverter.

CHAdemo and FC50 CCS are limited to 500 V [5]. When the voltage limit of a fast dc charger is lower than the battery voltage V_{bat} , the direct charging method becomes unavailable. To charge high-voltage batteries with existing low-voltage fast dc chargers, a dc–dc boost conversion is required. As detailed, the industry mainly demands double-voltage boosting.

To utilize existing widespread low-voltage fast dc chargers, onboard boost converters are widely added to the vehicle [1], [6]. However, attaching an onboard converter demands extra devices, which increases the weight and volume of EV. Therefore, adding a converter is an expensive and a driving range-reducing solution. Instead of adding additional converters, the motor-integrated charger topology has been introduced and commercialized. The integrated charger utilizes the traction system to charge the battery, where the stator windings function as inductors.

Various studies have explored integrated charger topologies and their control techniques. First, integrated charging methods utilizing ac power supply are studied. In [7], [8], [9], and [10], the rectifier output is connected to the neutral point of the motor and the negative side of the dc link, configuring a three-phase power factor correction (PFC) boost converter. In [11], an integrated PFC buck converter topology is presented, by reconfiguring the motor-inverter system using connectors. Next, dc–dc conversion techniques employing the traction system are studied. A charging method utilizing dc power supply for boost conversion is presented in [12] and [13], and the topology is illustrated in Fig. 1. As shown, this article particularly examines an integrated charger control strategy for setups without an intermediary converter between the battery and the inverter's dc link. Here, the fast dc charger ends are connected to the neutral point of the motor and the negative rail of the inverter. As shown, the integrated charger is structured as a three-phase

interleaved boost converter. By utilizing this topology, in [13], control methods for vehicle-to-grid (V2G) buck and grid-to-vehicle (G2V) boost operations are studied. In [14], an integrated vehicle-to-vehicle (V2V) power transfer method is studied, connecting the integrated charger topology in a back-to-back formation. In [15], a charging method is proposed for a dual-source fed open-end winding motor, which simultaneously charges two batteries by a single fast dc charger.

Meanwhile, one of the main challenges of the integrated charger is that the efficiency does not reach that of additional onboard converters, those solely designed for charging. Motor losses are dominant, especially in motor-inverter systems, which mainly consist of conduction losses and core losses [16], [17]. Such dominance is also verified in motor-integrated chargers [18], [19]. Notably, reducing the ac component of the phase current diminishes ac conduction losses and core losses incurred in the motor [20], [21]. The conventional method uses low-zero-sequence inductance, which generates large current ripples and high-motor losses, especially when charging an 800 V battery with a 400 V fast dc charger. On the other hand, this article proposes a charging method that uses the differential-mode inductance and two-phase conduction, thereby reducing ac conduction losses and core losses. Meanwhile, since the charger current flows only into two windings, dc conduction losses increase. Considering both aspects, the proposed method improves the efficiency when the reduction in ac conduction losses and core losses exceeds the increment in dc conduction losses.

An essential aspect of integrated chargers is to keep the rotor at a standstill during charging. Since the stator windings are utilized, each phase current must be regulated not to produce dc torque. Otherwise, nonzero torque causes parking gear or brake stress. Moreover, when failures occur in the parking devices, nonzero torque may lead to unintended rotation, causing safety issues. To prevent rotation, a charging strategy that controls the dc value of each phase current the same has been generally adopted in the conventional studies and the industry [9], [11], [12], [13], [14], [15]. Moreover, in [11], it is emphasized that the only condition to suppress rotation, regardless of the rotor's position θ_r , is to regulate each dc current the same. In the proposed method, on the other hand, each current is differently regulated according to θ_r , to ensure zero-torque while only two-phase currents are utilized.

In comparison with the conventional three-phase charging methods, the proposed two-phase charging technique enhances the efficiency. First, the effectiveness of the two-phase charging method is elaborated by inductance analysis and loss measurement results. Next, the strategy for selecting two phases and determining the current values to prevent the rotation is examined. Finally, the laboratory experiment is shown to verify the effectiveness of the proposed method.

The main contributions of this article are as follows.

- 1) A two-phase charging method is proposed to increase the efficiency of the integrated charger, differing from the conventional three-phase approach.
- 2) Current ripple is analyzed based on the inductance of the integrated charger, and its impact on losses is evaluated.

The proposed method uses differential-mode inductance to reduce the ripple and increase the efficiency.

- 3) The zero-torque condition is analyzed. The proposed method not only suppresses torque but also improves positional stability by position control.

II. INDUCTANCE ANALYSIS AND LOSS MEASUREMENT OF INTEGRATED CHARGER

This research proposes a charging method that enhances the efficiency of the integrated charger by reducing motor losses. Notably, motor losses predominantly consist of conduction losses and core losses. Within these, ac conduction losses and core losses decrease as the amplitudes of alternating stator currents reduce. In this section, a charging method that lowers the overall losses by reducing current ripples is proposed. First, the inductance of integrated charger is analyzed to study the current response of different charging methods. Next, loss measurement results corresponding to the charging methods are presented.

A. Analysis of Integrated Charger Inductance

The magnetic flux linkage in a , b , and c phases λ_{abcs} and the stator inductance L_{abcs} satisfy the following:

$$\lambda_{abcs} = L_{abcs} i_{abcs} + \begin{bmatrix} L_{ns} \\ L_{ns} \\ L_{ns} \end{bmatrix} i_{ns} + \begin{bmatrix} \cos \theta_r \\ \cos \left(\theta_r - \frac{2\pi}{3} \right) \\ \cos \left(\theta_r + \frac{2\pi}{3} \right) \end{bmatrix} \psi_f \quad (1)$$

$$L_{abcs} = L_{ls} \mathbf{I} + \frac{L_d + L_q - 2L_{ls}}{6} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} + \frac{L_d - L_q}{3} \begin{bmatrix} \cos 2\theta_r & \cos 2\left(\theta_r - \frac{\pi}{3} \right) & \cos 2\left(\theta_r + \frac{\pi}{3} \right) \\ \cos 2\left(\theta_r - \frac{\pi}{3} \right) & \cos 2\left(\theta_r + \frac{\pi}{3} \right) & \cos 2\theta_r \\ \cos 2\left(\theta_r + \frac{\pi}{3} \right) & \cos 2\theta_r & \cos 2\left(\theta_r - \frac{\pi}{3} \right) \end{bmatrix} \quad (2)$$

where i_{abcs} is the a , b , and c phases stator currents, L_{ns} is the zero-sequence inductance of the motor, $\mathbf{I}$ is the 3-by-3 identity matrix, i_{ns} is the zero-sequence current (ZSC), ψ_f is the permanent magnet flux linkage, L_{ls} is the stator leakage inductance, and L_d and L_q are the d - and q -axis inductance of the stator windings. For fixed θ_r , the stator phase voltage v_{abcs} is represented as

$$v_{abcs} = R_s i_{abcs} + \frac{d\lambda_{abcs}}{dt} \cong \frac{d\lambda_{abcs}}{dt} = \left(L_{abcs} + \frac{L_{ns}}{3} \mathbf{J}_3 \right) \frac{di_{abcs}}{dt} = L_{abcs}^* \frac{di_{abcs}}{dt} \quad (3)$$

where R_s is the stator resistance, and $\mathbf{J}_3$ is a 3-by-3 all-ones matrix. For L_{abcs}^* , the eigenvectors $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ and the corresponding eigenvalues k_1 , k_2 , and k_3 are derived as

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} \cos(\theta_r) \\ \cos\left(\theta_r - \frac{2\pi}{3}\right) \\ \cos\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix} \sin(\theta_r) \\ \sin\left(\theta_r - \frac{2\pi}{3}\right) \\ \sin\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix}$$

$$k_1 = L_{ls} + L_{ns}, k_2 = L_d, k_3 = L_q. \quad (4)$$

Applying Park transformation, $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ corresponds to unit vectors in the direction of n -, d -, and q -axis, respectively. The eigenvectors are linearly independent and span $\mathbb{R}^3$. Hence, $\mathbf{v}_{abc}$ and $\mathbf{i}_{abc}$ are separable into zero-sequence and differential-mode components as

$$\mathbf{v}_{abc} = v_{ns}\mathbf{x}_1 + v_{DMd}\mathbf{x}_2 + v_{DMq}\mathbf{x}_3 \quad (5)$$

$$\mathbf{i}_{abc} = i_{ns}\mathbf{x}_1 + i_{DMd}\mathbf{x}_2 + i_{DMq}\mathbf{x}_3 \quad (6)$$

where v_{ns} , v_{DMd} , and v_{DMq} denote the sizes of the voltage components in n -, d -, and q -axis directions, respectively. Here, v_{ns} is also termed as the zero-sequence voltage (ZSV) and is equal to $(v_{as} + v_{bs} + v_{cs})/3$. Substituting (5) and (6) into (3), the following equations are derived as:

$$\begin{cases} v_{ns} = (L_{ls} + L_{ns}) \frac{di_{ns}}{dt} \\ v_{DMd} = L_d \frac{di_{DMd}}{dt} \\ v_{DMq} = L_q \frac{di_{DMq}}{dt} \end{cases}. \quad (7)$$

As presented, different inductances are effective depending on the components of applied voltage. The zero-sequence inductance of the integrated charger L_{ZS} , is $L_{ls} + L_{ns}$, and the differential-mode inductances to voltage vectors parallel with d - and q -axis are L_d and L_q , respectively. Generally, L_d and L_q are significantly larger than $L_{ls} + L_{ns}$ in interior permanent magnet synchronous motors (IPMSMs); hence, the differential-mode inductance is greater than L_{ZS} . Because the time derivative of the current is inversely proportional to the inductance for a given applied voltage, the phase current ripple is predominantly attributed to v_{ns} . Since the motor losses are predominantly affected by the alternating currents, reducing $|v_{ns}|$ decreases ac conduction losses and the core losses, thereby enhancing the efficiency.

B. Loss Measurement Results in Zero-Sequence and Differential-Mode Responses

The fluctuations in phase currents generate ac conduction losses and core losses, which take significant portions in the motor losses. Here, the loss measurement results are presented when the zero-sequence and the differential-mode voltages are applied. The experimental parameters are listed in Table I. To mirror the practical applications requiring double voltage dc-dc conversion, V_{bat} and V_{ch} are scaled-down to 400 and 200 V, respectively.

In the boost converter depicted in Fig. 1, the ideal duty ratio of the high-side switches d_H is equal to V_{ch}/V_{bat} , which is the inverse of G_v . Hence, in a double-voltage setup, d_H reaches 0.5.

TABLE I
EXPERIMENTAL PARAMETERS

Motor Parameters	Symbol	Value
Rated power	P_{rated}	3 kW
d -axis stator inductance	L_d	4.54 mH
q -axis stator inductance	L_q	7.66 mH
Stator leakage inductance	L_{ls}	0.135 mH
Stator resistance	R_s	0.358 Ω
Permanent magnet flux linkage	ψ_f	0.079 V · s/rad
System Parameters	Symbol	Value
DC link (battery) voltage	V_{bat}	400 V
Fast dc charger output voltage	V_{ch}	200 V
Switching frequency	f_{sw}	10 kHz

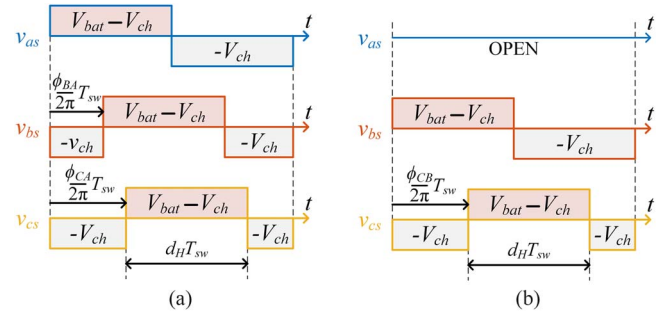


Fig. 2. Ideal phase voltage waveforms of the integrated charger in a switching period T_{sw} when (a) three-phase charging method and (b) two-phase charging method utilizing b and c phases are applied.

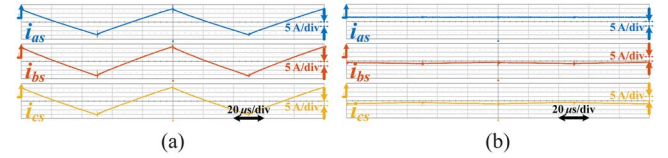


Fig. 3. Current waveforms of the integrated charger, f_{sw} is set to 10 kHz. When (a) only zero-sequence voltages are applied by IC3 method and (b) only differential-mode voltages are applied by SC2 method.

Fig. 2(a) illustrates the ideal phase voltage waveforms in steady state when all three phases are utilized for charging. ϕ , the phase shift angle between phases alters the voltage waveforms, which, in turn, determine the current waveforms. When in-phase switching is applied ($\phi_{BA} = \phi_{CA} = 0$), the phase voltages contain solely the zero-sequence component. This technique is identified as the three-phase charging method with in-phase carrier switching (IC3). The steady-state current waveforms of integrated charger at rated power ($P_{ch} = 3$ kW) are presented in Fig. 3(a), where the dc components of phase current (I_{as}, I_{bs}, I_{cs}) are controlled to $(-5, -5, -5)$ A. In the IC3 mode, the instantaneous voltage values of (v_{as}, v_{bs}, v_{cs}) are either $(200, 200, 200)$ V or $(-200, -200, -200)$ V. Applying the Park transformation, it equates to $\vec{v}_{dqms} = (v_{ds}, v_{qs}, v_{ns})$ being $(0, 0, 200)$ V or $(0, 0, -200)$ V. Since only v_{ns} is applied at any instant, the resulting current responses are determined by L_{ZS} .

Experiments are also conducted when inverter switching signals are applied only to b and c phases, as shown in Fig. 2(b).

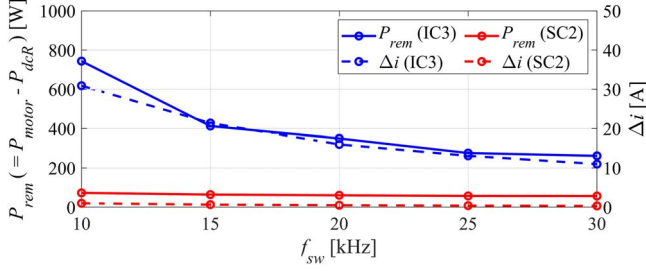


Fig. 4. P_{rem} and Δi when IC3 and SC2 methods are adopted, as f_{sw} varies from 10 to 30 kHz.

Implementing interleaved switching on these phases performs two-phase charging with 180°-shifted carrier switching (SC2). Referring to Fig. 2(b), as d_H equals to 0.5 under the double-voltage setup, voltages with an equal magnitude but in opposite sign are applied to b and c phases when SC2 method is utilized ($\phi_{CB} = \pi$). Moreover, referring to (3), when opposite voltages are applied to two phases while the remaining phase is opened, a negligible magnitude of voltage is applied to the remainder. Consequently, (v_{as}, v_{bs}, v_{cs}) of $(0, 200, -200)$ V or $(0, -200, 200)$ V are applied when SC2 method is used for phases b and c . Fig. 3(b) demonstrates the current waveforms as the SC2 method controls (I_{as}, I_{bs}, I_{cs}) to $(0, -7.38, -7.62)$ A. This set of current orients θ_r to zero, which is found by the experiment. The set of currents that does not induce rotation will be discussed further in Section III. At $\theta_r = 0$, $\vec{v}_{dqns}$ applied by the SC2 method is either $(0, 400/\sqrt{3}, 0)$ V or $(0, -400/\sqrt{3}, 0)$ V. Since the voltage vectors are parallel to q -axis, L_q determines the current response.

Fig. 3 demonstrates that utilizing the SC2 method reduces current ripple compared to the IC3 method. As analyzed, the average value of peak-to-peak current Δi is significantly reduced as $|v_{ns}|$ being suppressed by the SC2 method. When adopting SC2 method, differential-mode inductance is effective, hence Δi reduces. Conversely, when $|v_{ns}|$ is applied by IC3, Δi is noticeably increased because L_{ZS} is effective in this case.

As f_{sw} changes from 10 to 30 kHz, the experimental results for both methods are illustrated in Fig. 4. P_{rem} is termed as the remaining motor losses, obtained after the subtraction of dc conduction losses P_{dcR} from the motor losses P_{motor} . Here, P_{dcR} and P_{rem} are expressed as

$$P_{dcR} = \sum_{x=a,b,c} R_s I_{xs}^2 \quad (8)$$

$$P_{rem} = P_{motor} - P_{dcR} = P_{motor} - \sum_{x=a,b,c} R_s I_{xs}^2 \quad (9)$$

P_{rem} dominantly consists of losses induced by the current ripples. As demonstrated in Fig. 4, the SC2 method decreases P_{rem} as well as Δi . Moreover, an increase in f_{sw} also reduces Δi and P_{rem} . As shown, the reduction in P_{rem} is attributed to the reduction of current ripple. To conclude, diminishing $|v_{ns}|$ leads to a decrease in current ripples, thereby lowering motor losses and enhancing the efficiency of the integrated charger.

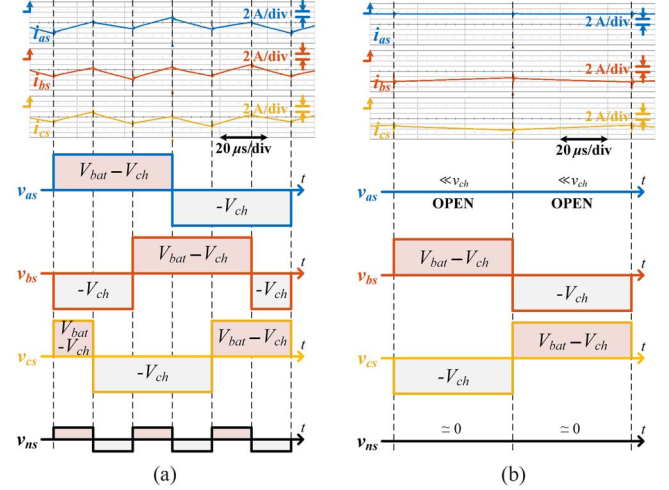


Fig. 5. i_{abcs} waveforms in the steady-state obtained by the experiment, illustrated with ideal v_{abcs} and v_{ns} . When (a) conventional SC3 method is utilized and (b) proposed SC2 technique is applied.

III. PROPOSED TWO-PHASE INTERLEAVED DC CHARGING STRATEGY

The previous section demonstrated that suppressing $|v_{ns}|$ for charging reduces motor losses induced by current ripples. For the next step, this section studies the strategy to reduce $|v_{ns}|$ at a voltage gain near 2 and proposes the current control method to ensure that no dc torque is generated during charging.

A. Proposal of Two-Phase Interleaved Charging Method

When identical switching signals are applied to all phases, the phase voltages contain solely the zero-sequence component. On the contrary, the phase shift between gating pulses suppresses $|v_{ns}|$ and thus reduces Δi , which is greatly decreased when the inverter operates as an interleaved boost converter. This technique is adopted in [11], [14], and [15], as a general switching method reducing current ripple. Here, three-phase interleaved switching is achieved by 120°-shifted carrier, which corresponds to $\{\phi_{BA}, \phi_{CA}\} = \{-2\pi/3, 2\pi/3\}$ in Fig. 2(a). This charging method is termed as the SC3 method.

Referring to Fig. 2(a), for a three-phase interleaved boost converter with d_H values of 1/3 or 2/3, which correspond to G_v of 3 or 1.5, perform the largest reduction in $|v_{ns}|$. In the same manner, at $d_H = 0.5$, indicating $G_v = 2$, $|v_{ns}|$ is maximally reduced by the SC2 method. Given the industrial demand for charging an 800 V battery with a 400 V fast dc charger, this article emphasizes the necessity of studying the SC2 technique.

Fig. 5(a) and 5(b) presents the waveforms when SC2 and SC3 methods are applied, respectively, at $G_v = 2$. The experiments are done in a rated power condition, where $I_{as} + I_{bs} + I_{cs} = -15$ A. The figures show that the SC2 method reduces $|v_{ns}|$ more effectively than the conventional SC3 technique in the double-voltage setup. As a result, Δi is notably decreased by SC2 method. Studies in [20] and [21] demonstrate that lowering the magnitude of Δi reduces core losses and ac conduction losses. As a result, the

proposed two-phase charging method effectively enhances the efficiency of the integrated charger, especially when the decrement of P_{rem} is larger than the increment of P_{dcR} . Loss measurement results have proven that SC2 method is more efficient in practical conditions, which are elaborated upon in the experimental results section.

B. Analysis of Zero Torque

First, the condition where the integrated charger does not produce dc torque is analyzed. In the charging mode, the sum of the phase currents is nonzero since the neutral point of the motor is utilized. As shown in Fig. 1, i_{ch} is separated into three phases, thus i_{ns} is derived as

$$i_{\text{ns}} = \frac{i_{\text{as}} + i_{\text{bs}} + i_{\text{cs}}}{3} = -\frac{i_{\text{ch}}}{3}. \quad (10)$$

Unlike the traction mode, the zero-sequence components exist and affect the net torque as

$$T_e = \frac{P_m}{\omega_r/pp} = \frac{e_{\text{as}}i_{\text{as}} + e_{\text{bs}}i_{\text{bs}} + e_{\text{cs}}i_{\text{cs}}}{\omega_r/pp} = 1.5 \frac{e_{\text{ds}}i_{\text{ds}} + e_{\text{qs}}i_{\text{qs}} + 2e_{\text{ns}}i_{\text{ns}}}{\omega_r/pp} \quad (11)$$

where P_m is the mechanical power of the motor, ω_r is the electrical angular speed, pp is the number of pole pairs, e_{dqn} is the dqn -axes back electromotive force (EMF), and i_{dqn} is the dqn -axes stator current in rotor reference frame [22], [23], [24]. Here, e_{dqn} is expressed as

$$\begin{cases} e_{\text{ds}} = -\omega_r L_q i_{\text{qs}} \\ e_{\text{qs}} = \omega_r L_d i_{\text{ds}} + \omega_r \psi_f \\ e_{\text{ns}} = \frac{d\psi_n}{dt} \end{cases} \quad (12)$$

where ψ_n is the zero-sequence magnetic flux linkage [25], [26], [27]. Substituting (12) into (11), T_e is rewritten as

$$T_e = 1.5pp\{\psi_f + (L_d - L_q)i_{\text{ds}}\}i_{\text{qs}} + 3pp\frac{\partial\psi_n}{\partial\theta_r}i_{\text{ns}}. \quad (13)$$

Depending on the contribution of each axis current, T_e is separated into T_{dq} and T_n as

$$T_{\text{dq}} = 1.5pp\{\psi_f + (L_d - L_q)i_{\text{ds}}\}i_{\text{qs}} \quad (14)$$

$$T_n = 3pp\frac{\partial\psi_n}{\partial\theta_r}i_{\text{ns}}. \quad (15)$$

It is noted that torque is also generated by i_{ns} and the angle-derivative of ψ_n . From (13), the zero-torque condition is derived as

$$\frac{T_e}{1.5pp(L_d - L_q)} = \left(i_{\text{ds}} - \frac{\psi_f}{L_q - L_d}\right)i_{\text{qs}} - \frac{2i_{\text{ns}}}{L_q - L_d}\frac{\partial\psi_n}{\partial\theta_r} = 0. \quad (16)$$

When 1) i_{ns} is determined by i_{ch} and 2) parameters ψ_f , L_d , and L_q are given, the set of zero-torque points form a hyperbola in the dq plane, as illustrated in Fig. 6. To maintain zero-torque and thus suppress rotation, regulation is required to direct the current vector in the dq plane $\vec{i}_{\text{dqs}}$ to the hyperbola.

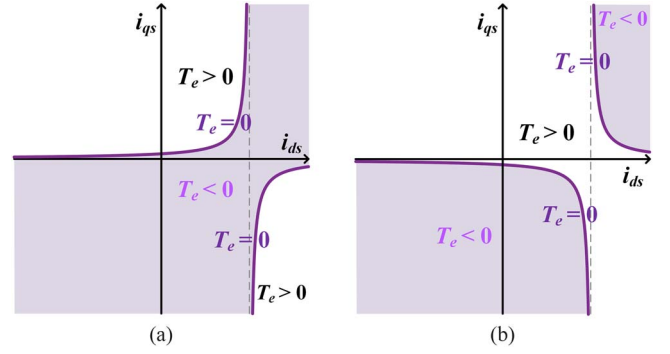


Fig. 6. dq plane and the sign of torque. (a) Sign of torque labeled in dq plane when $\partial\psi/\partial\theta_r$ is positive and (b) negative.

The conventional charging strategy controls the dc value of each phase current to be identical. Meanwhile, for a given fast dc charger current I_{ch} , $I_{\text{as}} = I_{\text{bs}} = I_{\text{cs}} = -I_{\text{ch}}/3$ corresponds to $I_{\text{ds}} = I_{\text{qs}} = 0$ and $I_{\text{ns}} = -I_{\text{ch}}/3$. From (14) and (15), the current vector $(I_{\text{ds}}, I_{\text{qs}}, I_{\text{ns}}) = (0, 0, -I_{\text{ch}}/3)$ suppresses T_{dq} , yet T_n is still present. In other words, T_n is generated by the conventional method. However, the resistive torque induced by parking gear and brake stress, and coulomb friction opposes it. Therefore, in practical applications, the conventional method keeps the motor at a standstill. On the other hand, with the proposed strategy, the current vector is directly regulated to the zero-torque hyperbola with unidentical phase current values. Consequently, the proposed method explicitly suppresses T_e by taking T_n into consideration.

C. Two-Phase DC Charging

This section analyzes the phase current combinations that satisfy $T_e = 0$ with only two-phase currents. Without loss of generality, let a and b phase currents charge the battery. Then, the following condition (17) is satisfied

$$\begin{cases} i_{\text{as}} + i_{\text{bs}} = -i_{\text{ch}} \\ i_{\text{cs}} = 0 \end{cases} \quad (i_{\text{as}}, i_{\text{bs}}, i_{\text{cs}} \leq 0). \quad (17)$$

The Park transformation is applied to express (17) in the stationary reference $\alpha\beta$ plane as

$$\begin{aligned} \begin{bmatrix} i_{\alpha s} \\ i_{\beta s} \end{bmatrix} &= \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ 0 & \sqrt{3} & -\sqrt{3} \end{bmatrix} \begin{bmatrix} i_{\text{as}} \\ i_{\text{bs}} \\ i_{\text{cs}} \end{bmatrix} \\ &= \frac{1}{3} \begin{bmatrix} -2i_{\text{ch}} - 3i_{\text{bs}} \\ \sqrt{3}i_{\text{bs}} \end{bmatrix}. \quad (i_{\text{bs}} \in [-i_{\text{ch}}, 0]). \end{aligned} \quad (18)$$

The set of $(i_{\alpha s}, i_{\beta s})$ from (18) corresponds to a linear function in the $\alpha\beta$ plane as

$$\begin{aligned} i_{\beta s} &= -\frac{1}{\sqrt{3}} \left(i_{\alpha s} + \frac{2}{3}i_{\text{ch}} \right) \\ &\cdot \left(i_{\alpha s} \in \left[-\frac{2i_{\text{ch}}}{3}, \frac{i_{\text{ch}}}{3} \right], i_{\beta s} \in \left[-\frac{i_{\text{ch}}}{\sqrt{3}}, 0 \right] \right). \end{aligned} \quad (19)$$

In the same way, distinct linear functions are derived for all three combinations of selecting two windings: a & b , b & c , and

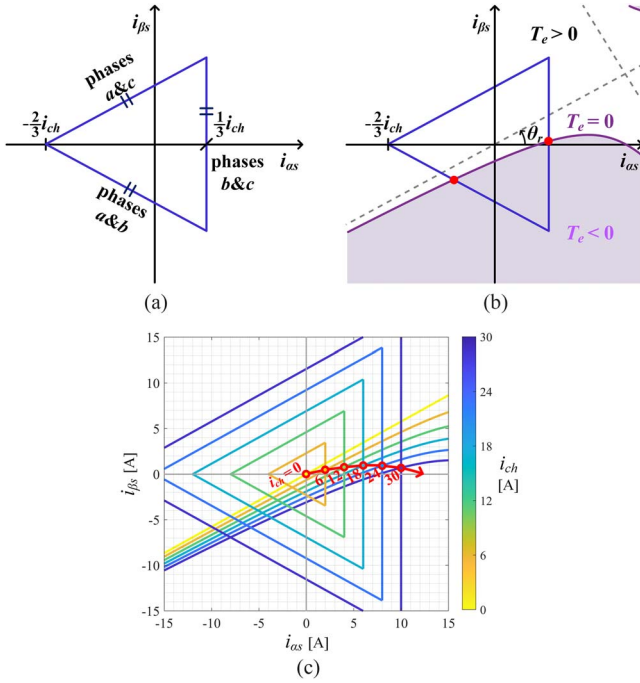


Fig. 7. Derivation of a TPZTC curve for given $\theta_r = \pi/6$. (a) Two-phase conduction points for given i_{ch} . (b) Two-phase conduction triangle and zero-torque hyperbola. The intersections are marked in red. (c) As i_{ch} varies, intersections form a TPZTC curve.

c & a . To sum up, the set of $(i_{\alpha s}, i_{\beta s})$ that satisfies two-phase conduction for a given i_{ch} forms an equilateral triangle in the $\alpha\beta$ plane, as shown in Fig. 7(a). Fig. 7(b) combines Figs. 7(a) and 6 into the same plot, highlighting the intersections in red; those achieved $T_e = 0$ with two-phase currents. Among the two existing intersections, the one closer to the center of the hyperbola is selected. For a given i_{ch} and θ_r , current regulation to the intersection ensures position stability, even when only two phases are utilized for charging.

As presented in Fig. 7(a), the two-phase excitation triangle expands as i_{ch} increases. In addition, as noted in (16) and Fig. 6, the gap between the two segments of a hyperbola widens as i_{ch} increases. Consequently, the intersection of the hyperbola and the triangle shifts as i_{ch} changes. With the variation of i_{ch} , the set of intersections forms a curve for a given θ_r , as illustrated in Fig. 7(c). In this article, this curve is named as the two-phase zero-torque charging (TPZTC) curve. To ensure $T_e = 0$, the reference of each phase current is determined according to the TPZTC curve, as i_{ch} varies. In conclusion, when $\vec{i}_{\alpha\beta s}$ is controlled along the TPZTC curve, the rotation is suppressed with only two-phase currents.

IV. CONTROL METHOD

In scenarios when the parking gear and brake malfunction, the rotor may slightly rotate even by small amount of torque, which can cause safety issues. To prevent such problems, a control method that eliminates dc torque is necessary. As previously analyzed, in two-phase charging method, $\vec{i}_{\alpha\beta s}$ must be controlled along the TPZTC curve, which is determined by the

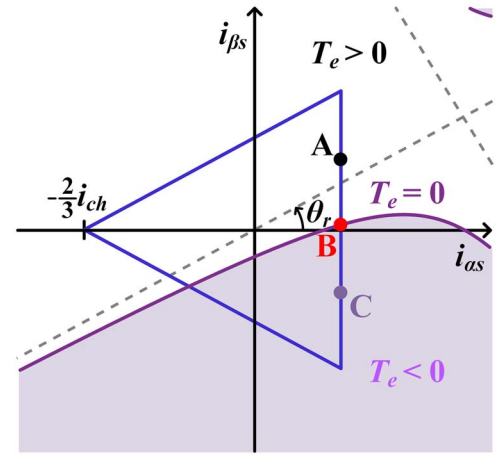


Fig. 8. Two-phase conduction points and the sign of torque. Including the point B marked in red, three points are illustrated on the triangle.

motor parameters. However, these parameters change with variations in temperature θ_r and dqn -axes currents. A practical approach for reflecting the variations to the TPZTC curve is utilizing a look-up table (LUT) obtained by experiments or finite element analysis (FEA). Yet, reflecting factors such as temperature into LUT is challenging. For this reason, achieving zero-torque by open-loop control is difficult. In this section, a position controller that maintains the rotor at a standstill is proposed. The controller explicitly suppresses the rotation, even without LUT of the TPZTC curve. By incorporating position control, stability in θ_r is reinforced; thus, a multibrake system is established.

Fig. 8 illustrates the sign of torque on the $\alpha\beta$ plane for a given θ_r and i_{ch} . The intersection of the two-phase conduction triangle and the zero-torque hyperbola is marked in red (point B), denoting the optimal point achieving two-phase charging and zero-torque at the same time. For an arbitrary current vector $\vec{i}_{\alpha\beta s}$ in the $\alpha\beta$ plane, θ_i is defined as

$$\theta_i = \tan^{-1}(i_{\beta s}/i_{\alpha s}). \quad (20)$$

θ_i represents the counterclockwise angle of $\vec{i}_{\alpha\beta s}$ from the α -axis. Increasing θ_i from point B (moving to point A) results in positive torque, while decreasing it (moving to point C) generates negative torque. These characteristics allow θ_i to regulate the torque in the position controller. Therefore, $\vec{i}_{\alpha\beta s}$ must be regulated to control θ_r in the charging process.

The structure of the controller is demonstrated in Fig. 9. Ahead of the charging procedure, the initial rotor position is set as the position reference, θ_r^* . From θ_r^* and sensed θ_r , the position controller generates θ_i^* , the θ_i reference, instead of torque reference T_e^* . Next, a , b , and c phase current references $\vec{i}_{abc s}^*$ are calculated by θ_i^* and the two-phase conduction triangle, for a given fast dc charger current reference i_{ch}^* . Finally, this control method leads to a phase current combination that uses only two-phase currents for charging while maintaining the initial rotor position.

Since minimizing charging time is the top priority, a charging method selector must be incorporated to choose the more

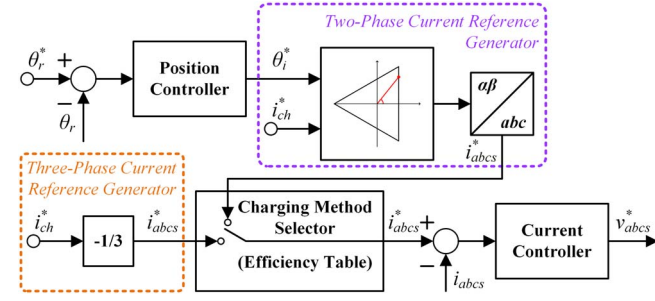


Fig. 9. Control block diagram of the integrated charger.

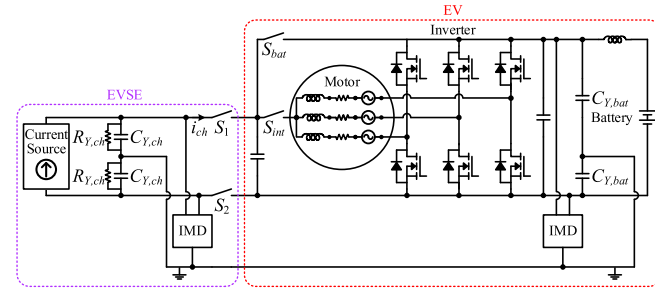


Fig. 10. DC charging system. Depending on the maximum voltage of the EVSE, either the direct charging or the integrated charging is selected.

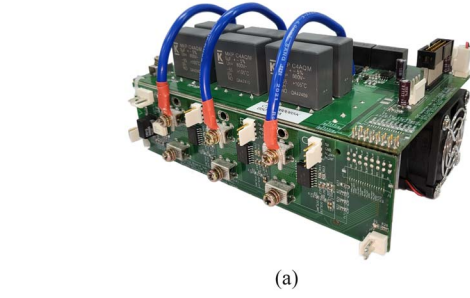
efficient method between the conventional SC3 and the proposed SC2 technique. This is because the conventional SC3 method may outperform the proposed one in the efficiency, especially when the increment of P_{dcR} is greater than the decrement of P_{rem} . If the efficiency table determines that the SC3 strategy is more efficient under the given conditions, then i_{as}^* , i_{bs}^* , and i_{cs}^* are all set to $-i_{ch}^*/3$.

V. CONSIDERATIONS FOR INTEGRATED CHARGER APPLICATION

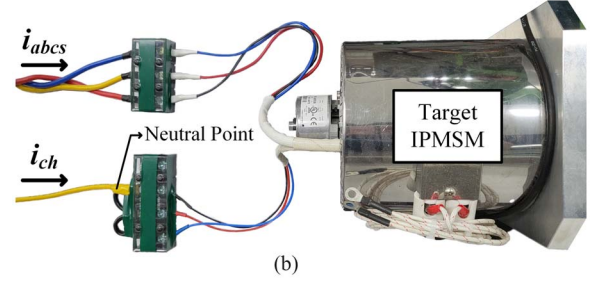
Connecting an integrated charger to an electric vehicle supply equipment (EVSE) forms a two-stage dc-dc converter. However, interactions between two different closed-loop regulated converters can lead to instability [28], [29]. To avoid that, the ratio of the EVSE's output impedance to the integrated charger's input impedance must meet the stability criterion. Beyond the fundamental Nyquist criterion, various criteria such as Middlebrook, GMPM, RESC, and ESAC should also be considered to ensure converter dynamics remain stable [30].

Moreover, protection measures are necessary to verify the safety of the charging process. Specifically, an appropriate isolation monitoring device (IMD) operation method must be established for integrated charger technology.

Fig. 10 illustrates a dc charging system. The vehicle chassis is connected to the EVSE ground, and charging begins by closing contactors S_1 and S_2 . S_{bat} must be closed for direct charging, or close S_{int} for integrated charger use. The standards for IMD operation are specified in IEC 61851-23. According to the guidelines, the EV-side IMD diagnoses isolation resistance before charging, while the EVSE-side IMD checks for isolation faults during



(a)



(b)

Fig. 11. Laboratory setup. (a) Three-phase voltage-source SiC inverter. (b) Target IPMSM and the current path. Here, one end of an open-end winding motor was shorted to be utilized as a neutral point.

charging [31]. These procedures also apply to integrated chargers, inspecting the isolation status of both the battery and the charger. The only difference from the direct charging method is that the battery voltage and charger voltage are different, requiring isolation resistance diagnostics for each voltage.

VI. EXPERIMENTAL RESULTS

The validity of the proposed charging method is verified by the experiment. Due to the voltage limit of the laboratory setup, the experiments are conducted in a scaled-down setting with $V_{bat} = 400$ V and $V_{ch} = 200$ V. The output power of the dc charger is controlled to the rated power of the motor ($P_{ch} = V_{ch}I_{ch} = P_{rated} = 3$ kW). The laboratory setup is presented in Fig. 11 and the parameters are listed in Table I. For the inverter, silicon carbide (SiC) MOSFETs (C3M0060065K by Wolf-speed) are used.

Fig. 12 shows the steady-state current waveforms when the conventional IC3 and SC3 methods and the proposed SC2 method are used, respectively. Here, θ_r is set to 0° . Since the conventional methods are incapable of position control, a load motor is utilized to fix θ_r to 0° . Conversely, the SC2 method conducts position control of $\theta_r = 0$, simultaneously with charging. Fig. 12(a) and 12(b) shows that the conventional charging techniques result in noticeable ripples. As studied, three-phase switching methods generate nonzero v_{ns} and thereby induce i_{ns} ripple, which is dominant in current ripples. On the contrary, as shown in Fig. 12(c), the two-phase interleaved switching suppresses v_{ns} in the double-voltage setup, thus reducing Δi .

Fig. 13 presents the loss measurement results when V_{bat} and control methods vary. The losses are distinguished from inverter losses P_{inv} , dc conduction motor losses P_{dcR} , and remaining motor losses P_{rem} . Here, P_{rem} is obtained by subtracting P_{dcR} from the motor losses. As analyzed in Section II, IC3 method

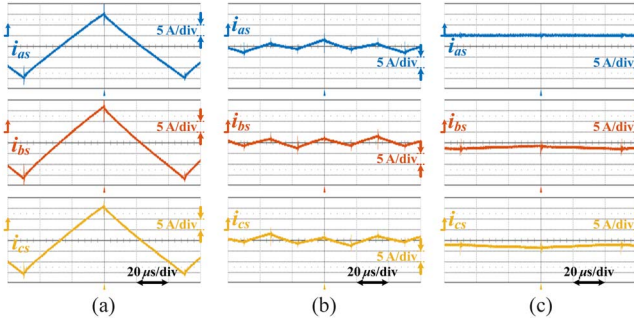


Fig. 12. Steady-state current waveforms obtained by experiments at $\theta_r = 0$, when: (a) in-phase carrier (IC3); (b) 120° phase-shifted carrier (SC3); and (c) proposed 180° phase-SC2 methods are utilized for charging.

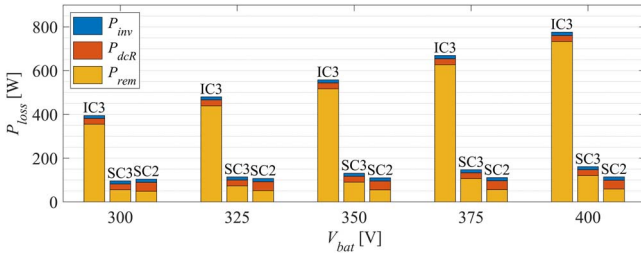


Fig. 13. Loss breakdown of the integrated charger, where $\theta_r = 0$. The losses are measured in three different charging methods as V_{bat} varies.

maximizes $|v_{ns}|$ and hence increases Δi and P_{rem} . On the other hand, SC3 and SC2 methods reduce $|v_{ns}|$ by interleaved switching and hence lower Δi and P_{rem} , thus the total losses are reduced. When comparing losses at $V_{bat} = 400$ V, applying SC2 method increases P_{dcR} and decreases P_{rem} , relative to the SC3 one. This is because the fast dc charger current flow through two phases in the SC2 method. Instead, the SC2 method reduces current ripple, thereby lowering AC losses. Since the decrease in P_{rem} is greater than the increase in P_{dcR} , the proposed SC2 method results in higher efficiency. This improvement is also observed when V_{bat} is greater than 300 V and is particularly effective as V_{bat} reaches 400 V. As explained in Section II-B, at $G_v = 2$, the SC2 method reduces $|v_{ns}|$, which in turn decreases P_{rem} . Conversely, as G_v approaches 1.5, the SC3 method effectively reduces total losses.

The results have validated that applying two-phase charging enhances the efficiency when charging a 800 V battery with a 400 V fast dc charger. Meanwhile, since V_{bat} varies according to the state of charge (SoC) of the battery, the most efficient charging method also vary. To minimize the charging time, the most efficient switching method must be selected, based on the efficiencies in SC2 and SC3 methods obtained as a function of V_{bat} .

Fig. 14 compares the charging efficiency and Δi of the SC3 and SC2 methods as f_{sw} alters from 10 to 60 kHz at rated power condition. As presented, the efficiency is enhanced by the proposed method across all experimented f_{sw} . As shown in Section II-B, charger losses are predominantly attributed to Δi ,

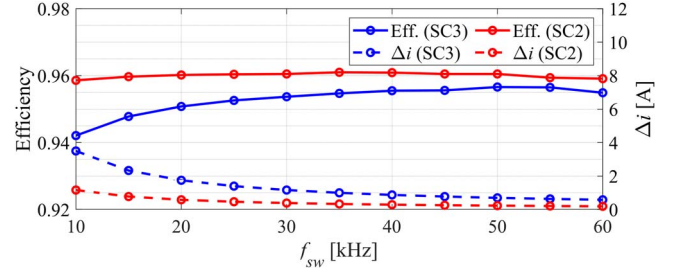


Fig. 14. Measured efficiency and Δi in two charging methods, where $\theta_r = 0$. The losses are measured as f_{sw} varies.

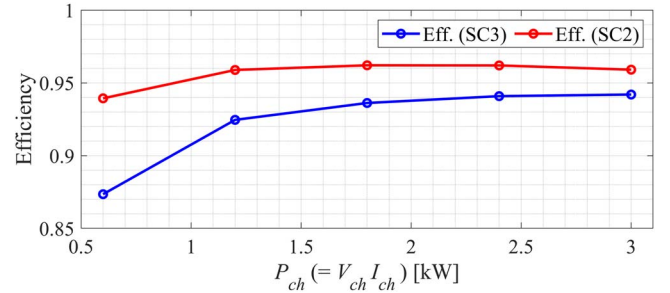


Fig. 15. Measured efficiency and Δi in two charging methods, where $\theta_r = 0$. The losses are measured as P_{ch} varies.

and this feature is also verified in Fig. 14. For both methods at lower frequencies, the efficiencies increase, and Δi decrease as f_{sw} elevates. However, for the SC3 method, the efficiency began to fall at 50 kHz, and for the SC2 method, the efficiency started to decline at 40 kHz. This is because a boost converter in continuous conduction mode (CCM) undergoes hard switching turn ON/OFF, resulting in switching losses. The switching losses increase almost linearly with f_{sw} , making P_{inv} dominant beyond 40–50 kHz and thus reducing the efficiency.

One of the main advantages of the proposed method is that it achieves high efficiency even in low f_{sw} range. The SC2 method achieved the efficiency of 95.9% even at 10 kHz, showing a 1.7% improvement from the conventional one. This result demonstrates that the proposed SC2 technique can achieve high efficiency, even if the inverter does not consist of wide bandgap switching devices.

Fig. 15 illustrates the charging efficiency from low power to the rated power. The efficiencies were obtained as I_{ch} increased from 3 to 15 A, with V_{bat} and V_{ch} fixed at 400 and 200 V, respectively. The results show that the proposed method achieved higher efficiency than the conventional method across all power ranges. At low I_{ch} levels, P_{dcR} takes a minor portion of the total loss, allowing the AC loss reduction by the proposed method to significantly enhance the efficiency. However, in the high I_{ch} range, the proportion of P_{dcR} increases, causing efficiency to decrease.

As discussed in Section III-C, when two-phase charging is utilized, the set of phase current that satisfy $T_e = 0$ varies with θ_r . This variation impacts P_{dcR} and therefore affects the efficiency. Fig. 16(a) presents the efficiency of SC3 and SC2

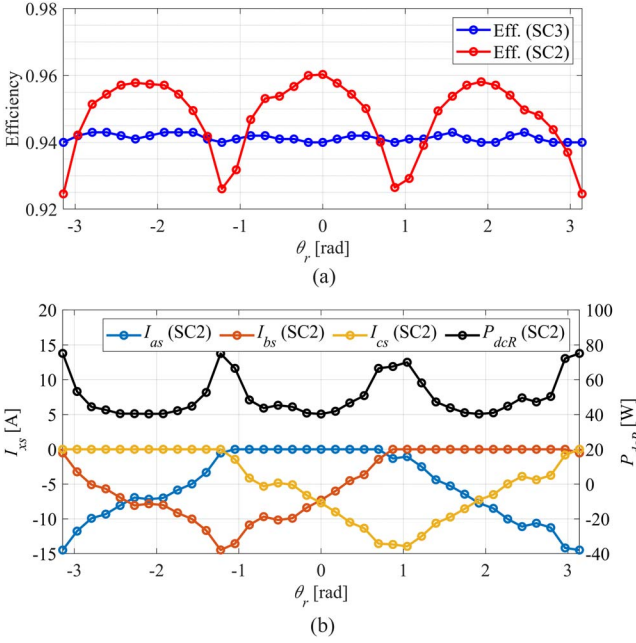


Fig. 16. Experimental results in two charging methods, as θ_r varies. (a) Efficiencies of SC3 and SC2 methods. (b) Phase current combinations those satisfy $T_e = 0$ in SC2 method and the corresponding P_{dcR} .

methods as θ_r changes, and Fig. 16(b) shows I_{abc} that satisfies $T_e = 0$ when SC2 method is utilized. The corresponding P_{dcR} at each I_{abc} is also presented. Experiments were conducted every 10 degrees. As shown in Fig. 16(b), between phases a , b , and c , current values that satisfy $T_e = 0$ demonstrate a 120° phase difference in relation to θ_r . This is due to the symmetrical distribution of the windings. Since P_{dcR} is proportional to the squared sum of the currents, P_{dcR} and the efficiency of SC2 method also exhibits a periodicity of 120°. Meanwhile, the conventional SC3 method exhibits similar efficiency levels, regardless of θ_r .

When utilizing the proposed charging method, P_{dcR} is minimized when two phase currents are regulated to the same dc value, according to the equality condition of the Cauchy–Schwarz inequality. When θ_r is around -120° , 0° , and 120° , high efficiencies are obtained since the two-phase currents are relatively similar in size. On the other hand, in ranges where P_{dcR} increases, the SC3 method may be more efficient than SC2. Yet, the proposed method is more efficient than the conventional one at majority of the angles. Alternatively, a strategy that adjusts θ_r to desired angles before charging is also feasible. By designing an automatic parking algorithm that aligns θ_r to angles where the proposed method is the most efficient, charging time is mostly reduced.

It is estimated that the efficiency would also be enhanced even when charging at industrial charger levels using EV motors. EV motors are designed with a lower R_s to enable high-power drive. For the EV motor used in [32], R_s is about 2% of that of the motor used in this experiment. Thus, even if the input

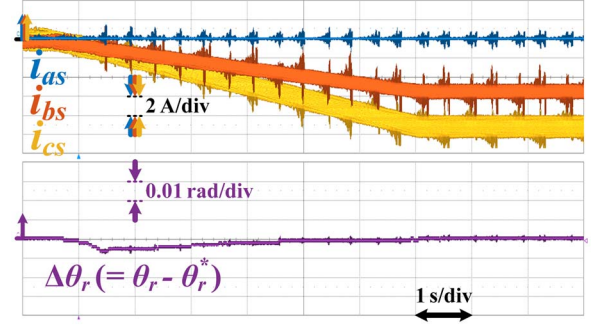


Fig. 17. Experimental results of $\Delta\theta_r$ as I_{ch} changes from 0 to 15 A, where $\theta_r^* = \pi/4$. The proposed two-phase charging method is utilized.

power increases from 30 to 150 kW with 50 times higher current, the proportion of dc conduction loss to the input power remains similar. Meanwhile, losses in EV motors mainly consist of core losses and AC conduction losses induced by AC current [17]. Since EV motors have lower inductance, current ripples are larger, leading to significant losses in EV motors having larger cores. Consequently, EV motors are highly compatible with the proposal since the proposed method enhances the efficiency by reducing AC losses which are dominant.

Fig. 17 demonstrates the rotor displacement angle $\Delta\theta_r = \theta_r - \theta_r^*$ and i_{abc} , as I_{ch} changes from 0 to 15 A when the proposed method is applied. This experiment is designed to verify the ability in controlling θ_r , as I_{ch} increases from 0 to the rated value. Notably, this examines the performance of rotor position control at the beginning of the charging. Here, $\Delta\theta_r$ is obtained by encoder. As shown in the figure, the proposed method effectively impedes rotation even though the charger current changes. Despite no load torque being applied, it was observed that θ_r deviated by less than 0.008 rad from θ_r^* . As a result of the position control, i_{abc} finally converges to the current combination that satisfies zero-torque and two-phase conduction at the same time.

VII. CONCLUSION

Recently, the battery voltage of EVs has doubled to 800 V. This has brought a challenge in using the conventional fast dc chargers, which are designed for conventional 400 V batteries. To deal with this matter, integrated charger topology has been studied. However, the conventional control method results in substantial motor losses. In this article, a two-phase charging method is proposed to reduce the losses. The proposed method reduces current ripple, thereby lowering motor losses. The impact of current ripple on motor losses is confirmed by loss measurement results. Moreover, zero-torque condition is analyzed to prevent rotation even when only two-phase currents are used for charging. A position controller is proposed to explicitly secure the rotor position. With the proposed control strategy, efficiencies are noticeably increased, and the rotor displacement is suppressed.

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